

Tableau

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What is Data Visualization?

- Data visualization is the representation of data through use of common graphics, such as charts, plots, infographics, and even animations.
- Benefits
 - Simplifies complex information
 - Identifies patterns and trends
 - Enhances decision-making

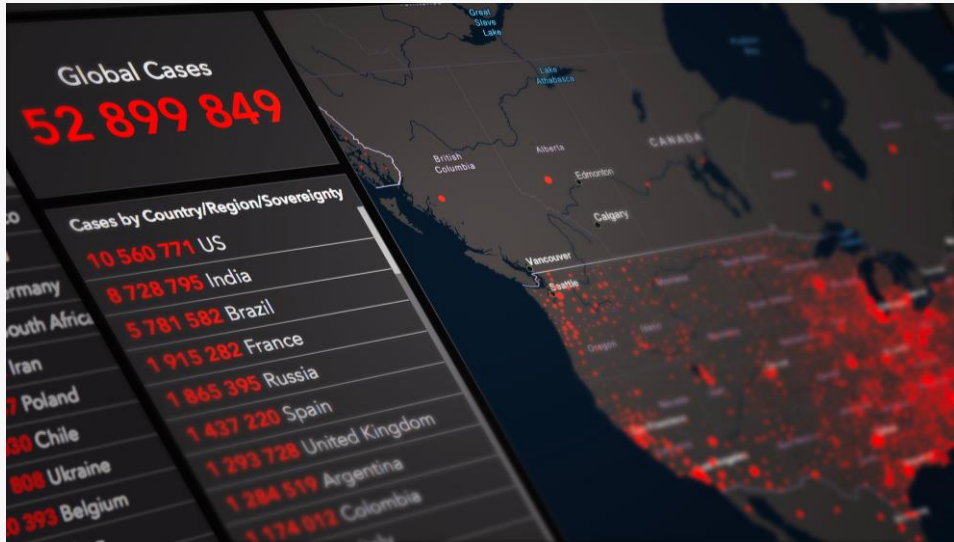




+tableau®

What is Tableau?

- Tableau is a powerful data visualization tool used to analyze, visualize, and present data
- It allows users to create interactive and dynamic dashboards, reports, and graphics



Key Features of Tableau

- Drag-and-drop functionality for easy data manipulation
- Wide variety of visualization options, including charts, graphs, maps, and more
- Ability to connect to various data sources, both structured and unstructured
- Interactive filters and parameters for real-time data exploration
- Tableau enables users to create visually appealing and interactive dashboards, charts, and graphs.

Benefits of Using Tableau

- Enhanced data-driven decision-making
- Quick and easy data analysis and visualization
- Improved data accuracy and quality
- Increased efficiency and productivity



Exploring Tableau's Connection Options

- Tableau offers versatile connectivity options to various data sources.
- 1. File Connectors: Connect to Excel, CSV, JSON, and other file formats.
- 2. Database Connectors: Connect to popular databases like MySQL, Oracle, and SQL Server.
- 3. Web Data Connectors: Access data from web services and APIs.
- 4. Cloud Connectors: Connect to cloud-based data sources like Google Analytics and Salesforce.
- 5. Big Data Connectors: Work with data stored in Hadoop, Spark, and other big data platforms.



Connect

To a File

Excel
Text file
Statistical file
More...

To a Server

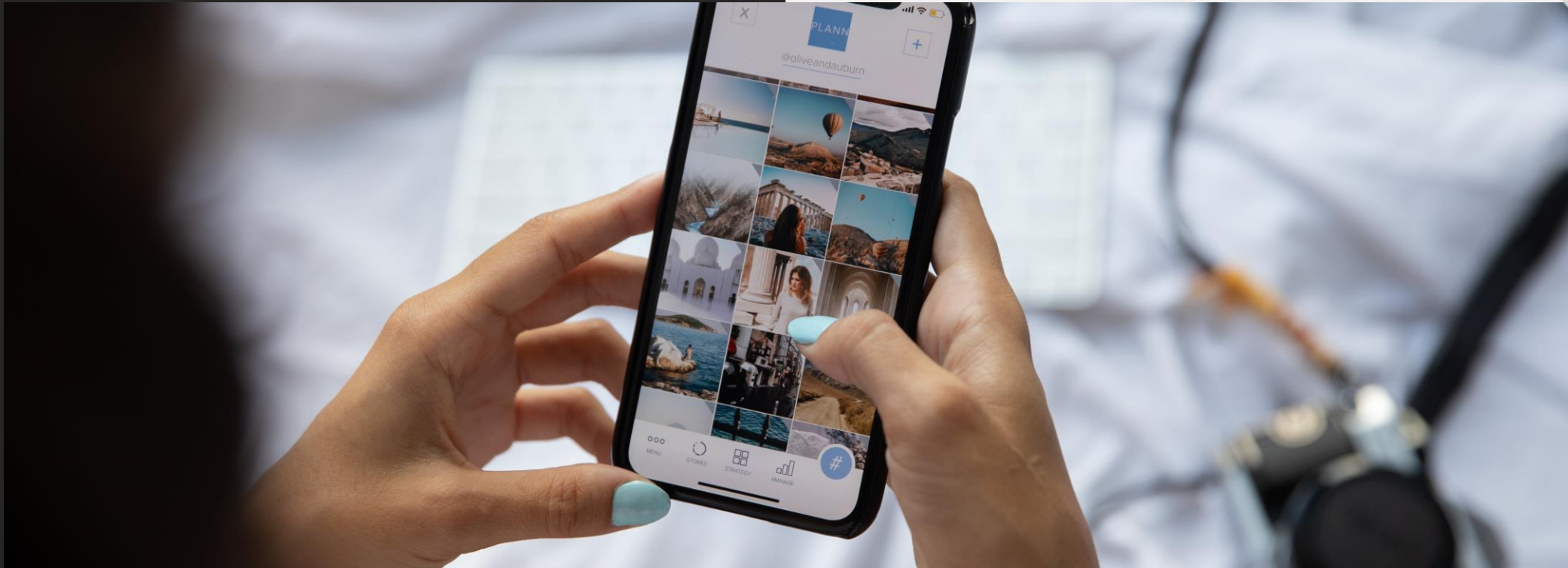
Tableau Server
Salesforce
Google Sheets
Microsoft SQL Server
MySQL
More... >

Saved Data Sources

Sample - Coffee Chain (csv)
Sample - Superstore - Englis...
Sample - Superstore Subset...
Sample - Superstore
Sample - World Bank Indicat...
Sheet1 (0315-ffl-list-washi...
Sheet27 (0215-ffl-list-wash...
Sheet46 (0115-ffl-list-wash...

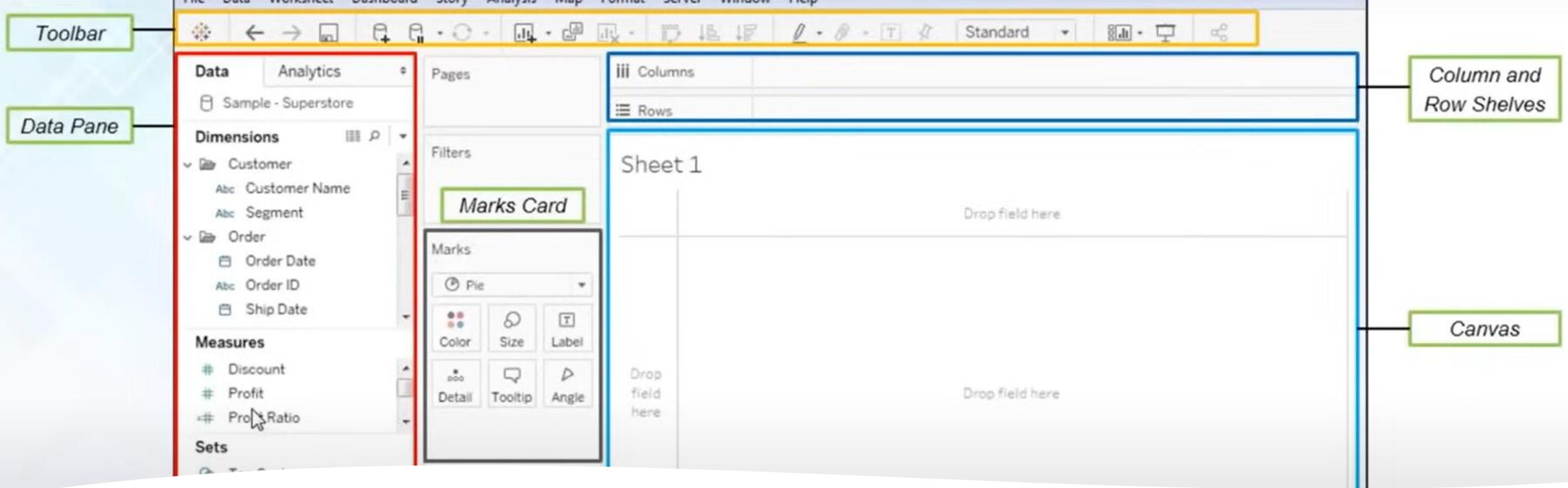


Tableau Server
Amazon Aurora
Amazon EMR
Amazon Redshift
Cloudera Hadoop
EXASolution
Firebird
Google Analytics
Google BigQuery
Google Cloud SQL
Google Sheets
Hortonworks Hadoop Hive
HP Vertica
Kognitio
MapR Hadoop Hive
MemSQL
Microsoft SQL Server
MySQL
OData
Oracle
Pivotal Greenplum Database
PostgreSQL
Presto
Snowflake
Spark SQL
Teradata
Web Data Connector



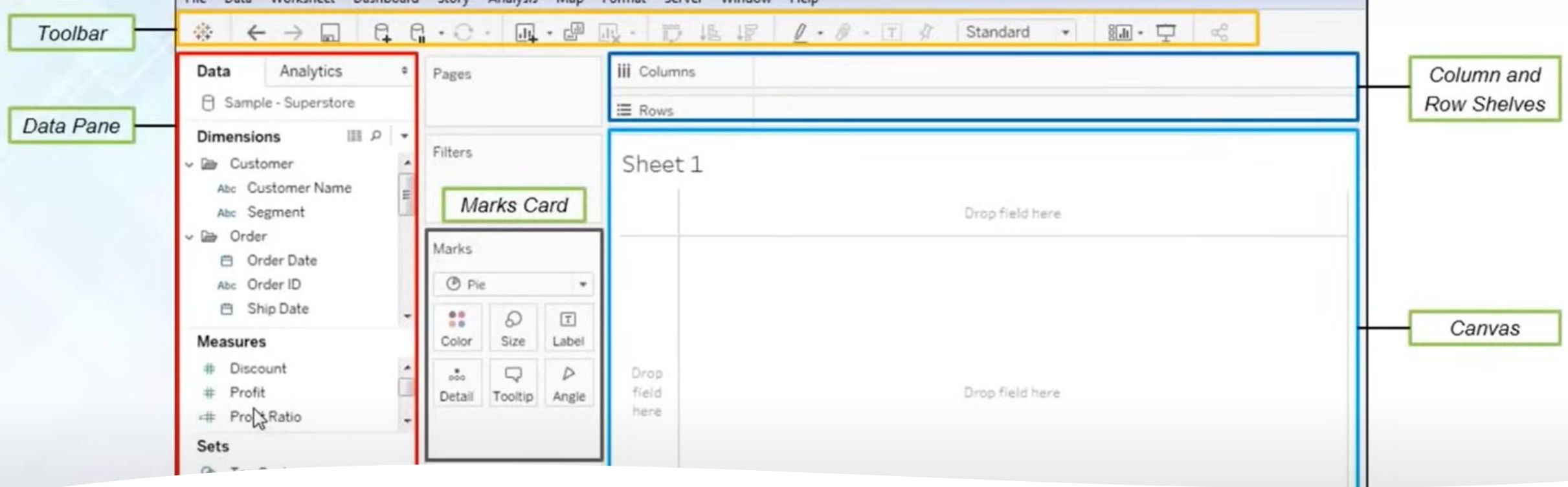
Understanding Tableau's User Interface

- Tableau is a powerful data visualization tool.
- The user interface consists of various elements that allow users to interact with the software.



Key Elements of Tableau's User Interface

- 1. Menu Bar: Provides access to various commands and tools.
- 2. Toolbar: Contains frequently used actions and shortcuts.
- 3. Data Pane: Displays the data sources and fields used in the workbook.
- 4. Worksheet: The main area where data visualization is created.
- 5. Sheets and Dashboards: Allows the organization and arrangement of multiple visualizations.



Key Elements of Tableau's User Interface

- 6. Show Me: Provides different visualization options based on the data.
- 7. Filters and Parameters: Enable data filtering and interactive control.
- 8. Legends and Color Palettes: Help interpret data and apply color schemes.
- 9. Analytics Pane: Offers various statistical and analytical functions.

Understanding Dimensions

- Dimensions in Tableau represent categorical data
- A dimension is a field that is an independent variable
- They provide context and structure to your analysis



Exploring Measures

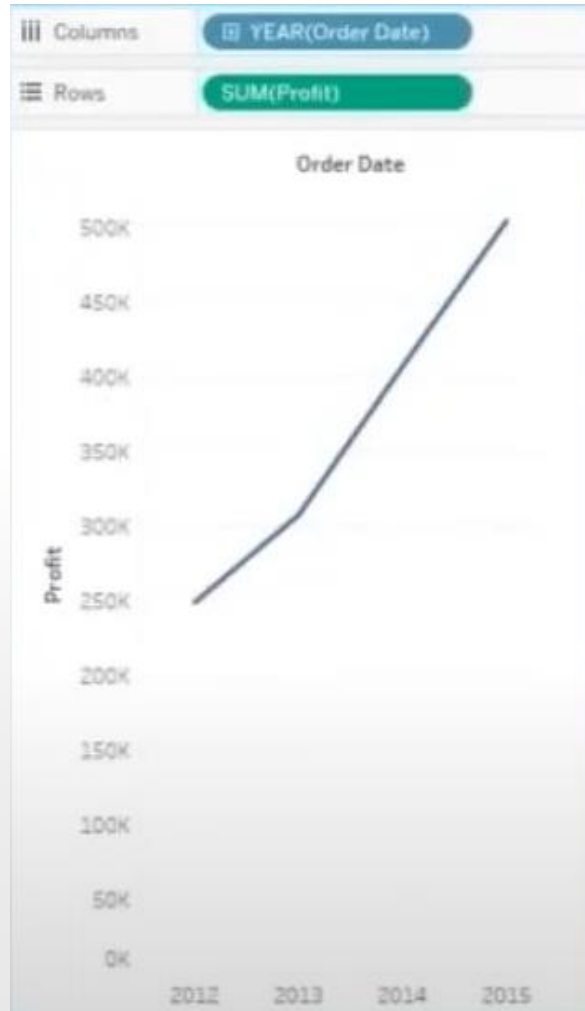
- Measures in Tableau represent numerical data
- It is a field that is a dependent variable and its value is a function of one or more dimensions.
- They allow you to perform calculations and aggregations



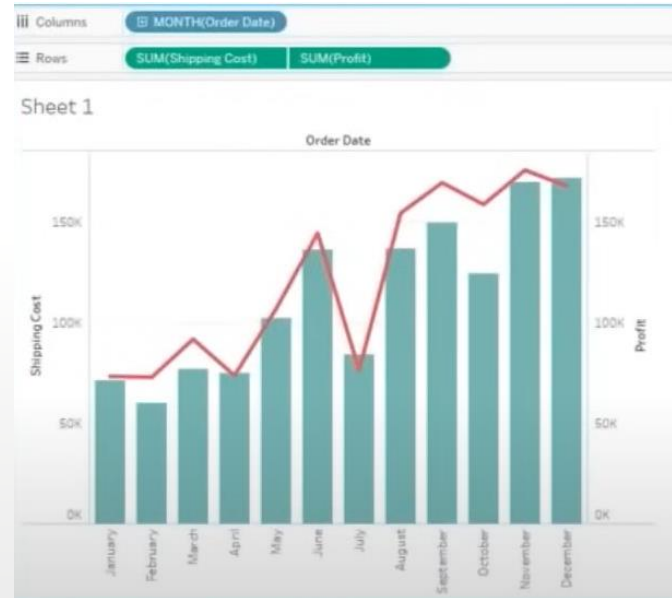


Data Visualizations

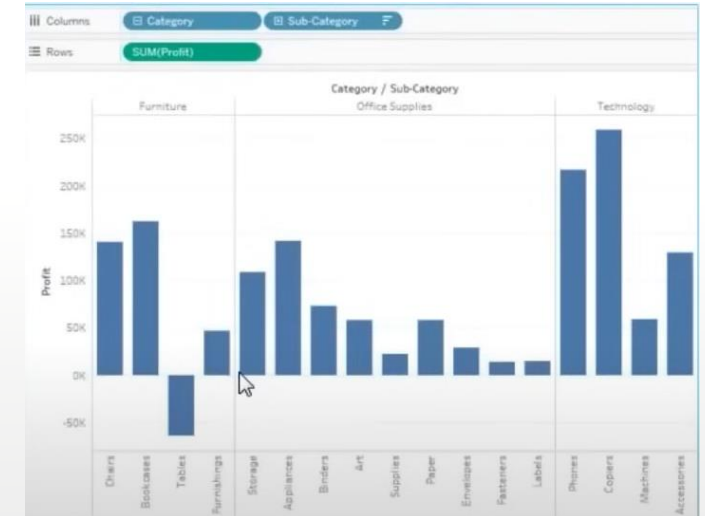
- Bar charts
- Line charts
- Pie charts
- Scatter plots
- Maps
- Heat maps
- Tree maps



Line Graph



Dual Axes Graph

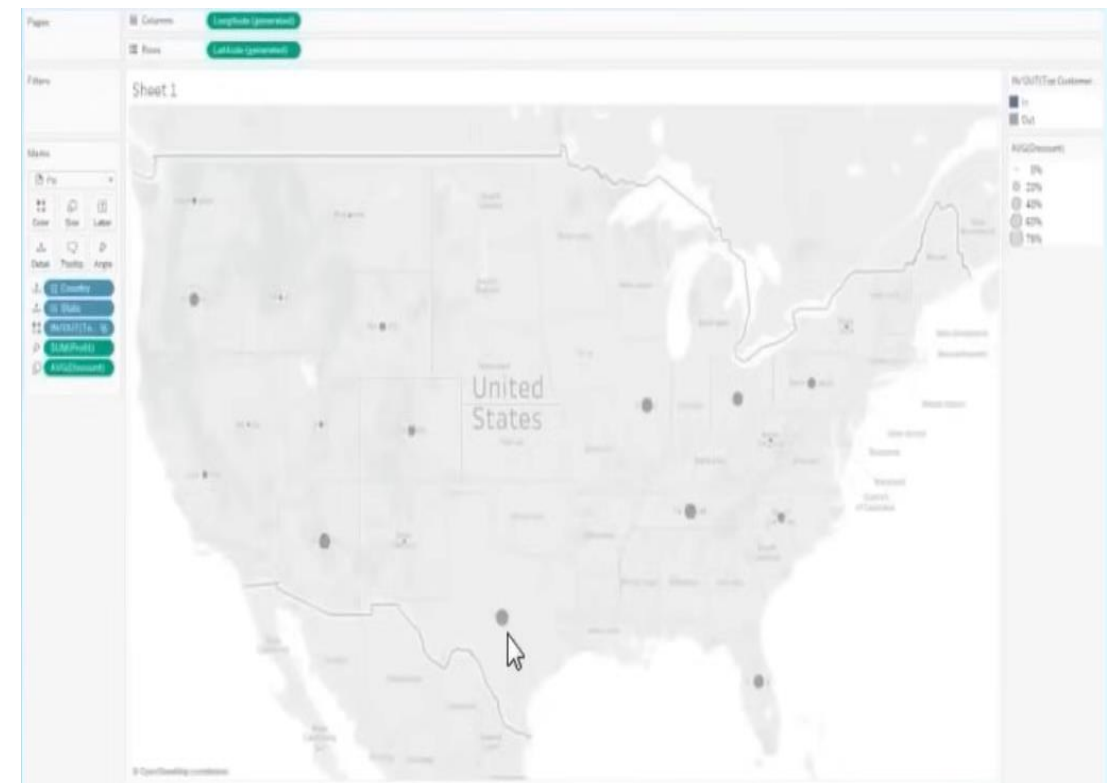


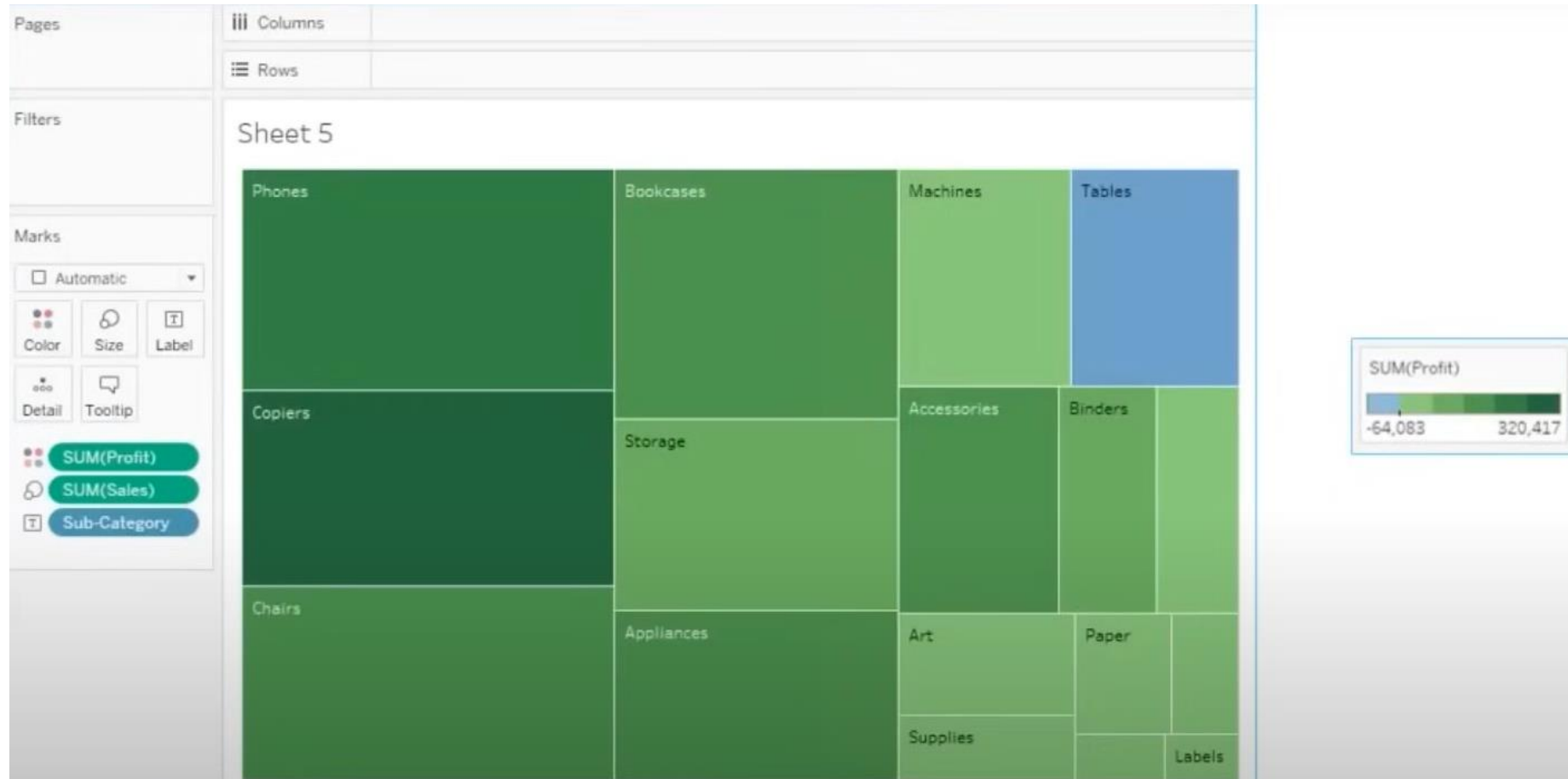
Bar Graph

Area Graph with Dual Axes

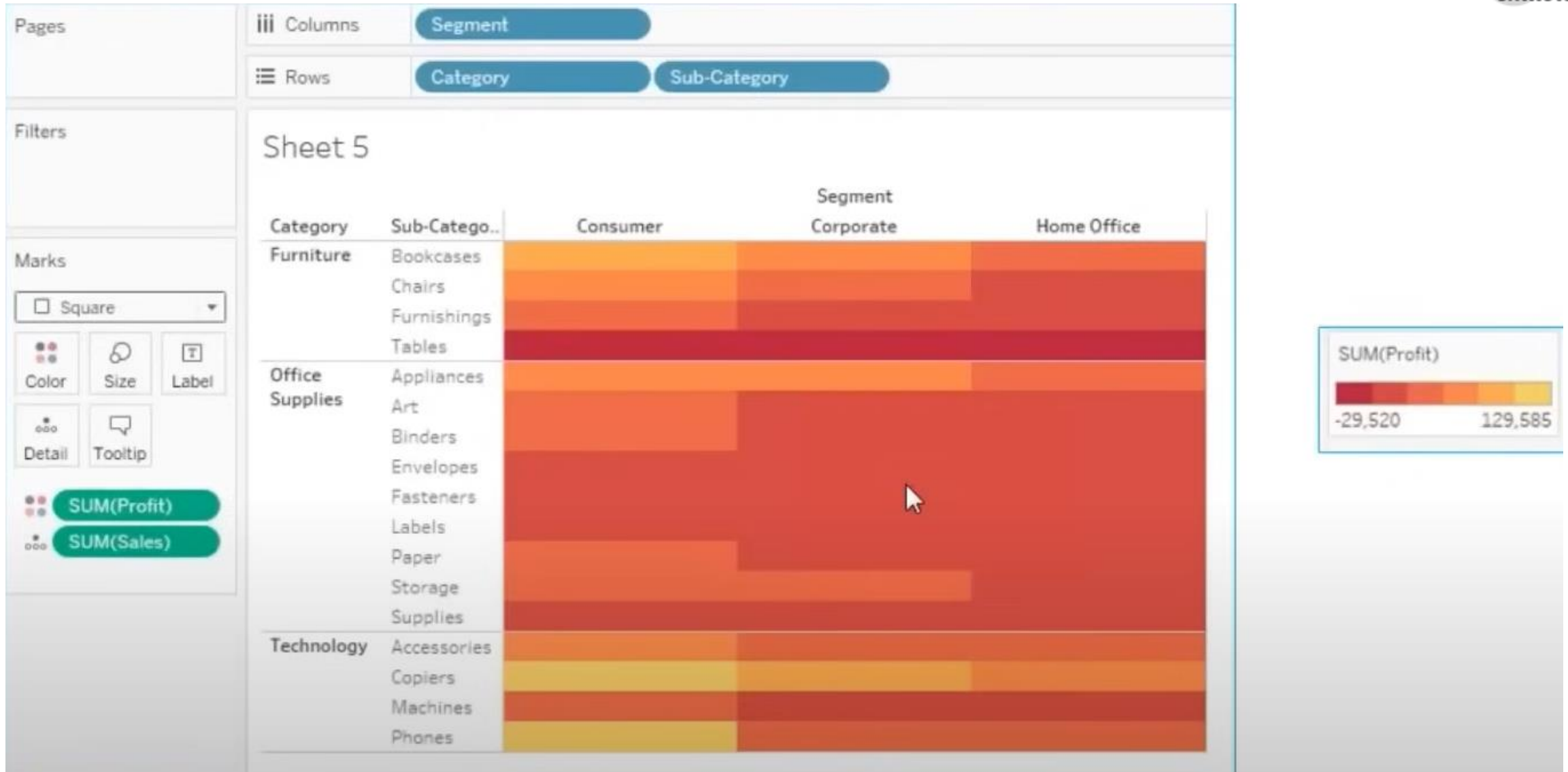


Geographical Graph

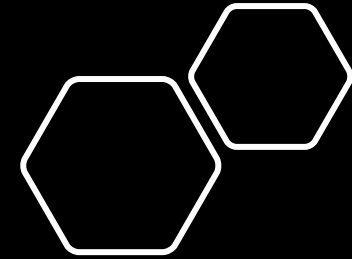




Tree map



Heat Map



VISUALIZATION	APPLICATION
Bar Graph	When dimension is continuous
Dual Axis Graph	For continuous dimensions
Line Graph	To represent two measures together
Area Graph (Dual Axes)	Provides better comparison amongst measures
Geographical graph	To plot measures on geographical map
Heat Map	To visualize variations between categories
Tree Map	To represent quantity in nested rectangles

Best Practices for Data Visualization

- Simplify and declutter
- Use appropriate color schemes
- Ensure data accuracy and quality
- Provide clear labels and titles
- Highlight key insights



Thank You



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kaggle

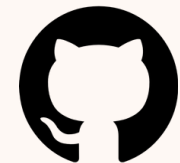
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Web Resources

<https://mitu.co.in>

<http://tusharkute.com>

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